Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 64. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the training loss.

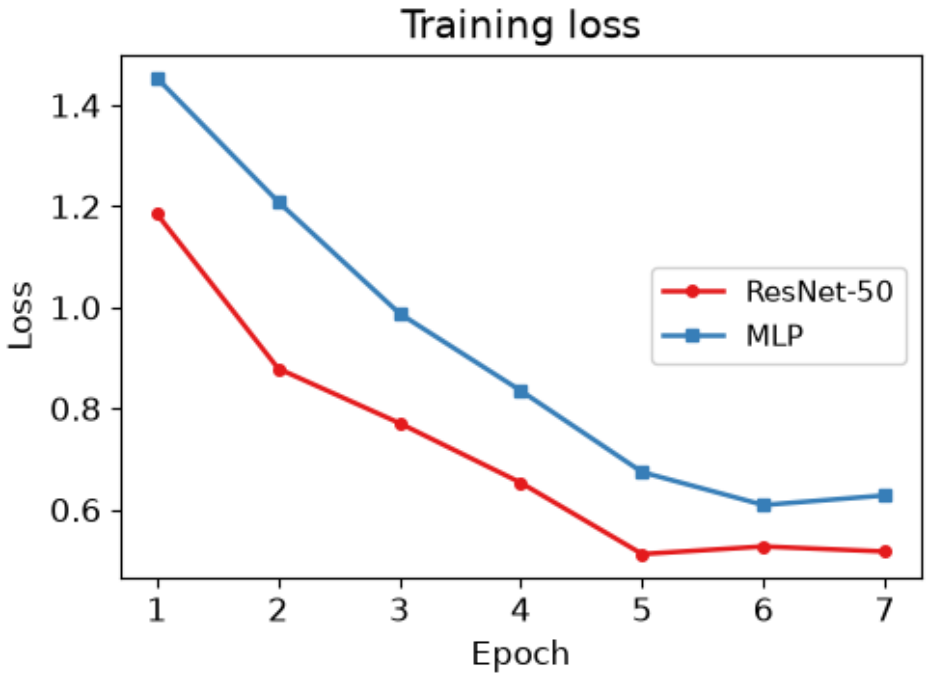


Figure 1: Training loss for the models in this experiment.

MLP has a loss of 1.454 at epoch 1. At epoch 4, the loss of ResNet-50 is 0.632 (Figure 1). As shown in Figure 1, ResNet-50 has a loss of about 0.5 at epoch 6.

Figure 2 shows the validation accuracy by dataset.

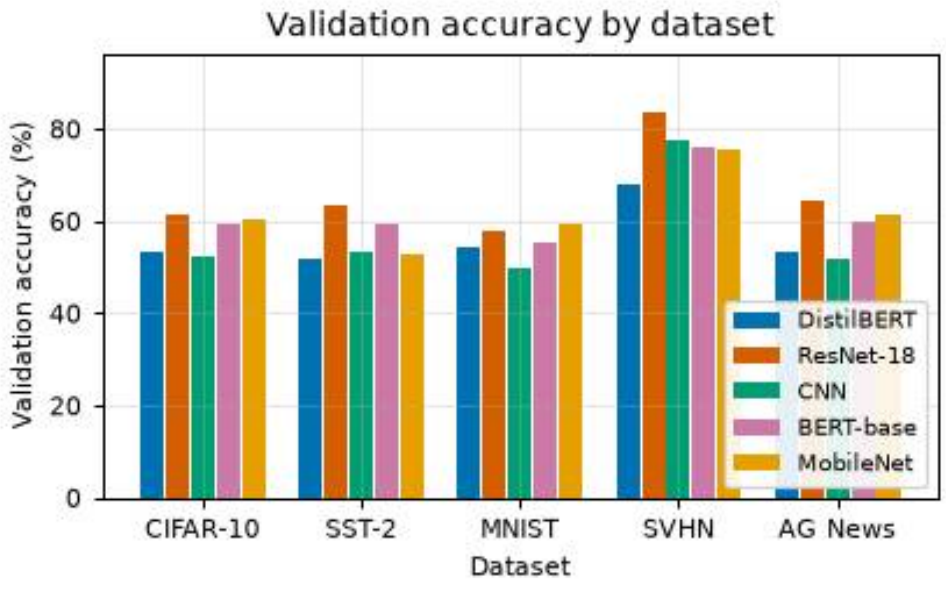


Figure 2: Validation accuracy by dataset for the models in this experiment.

On CIFAR-10, ResNet-18 obtains a validation accuracy of 61.4%. ResNet-18 outperforms BERT-base by

14.4 percentage points on AG News (Figure 2). As shown in Figure 2, BERT-base outperforms CNN by 54.2 percentage points on SST-2. As shown in Figure 2, on SVHN, ResNet-18 obtains a validation accuracy of 93.4%.

Figure 3 shows the accuracy of BERT-base by dataset.

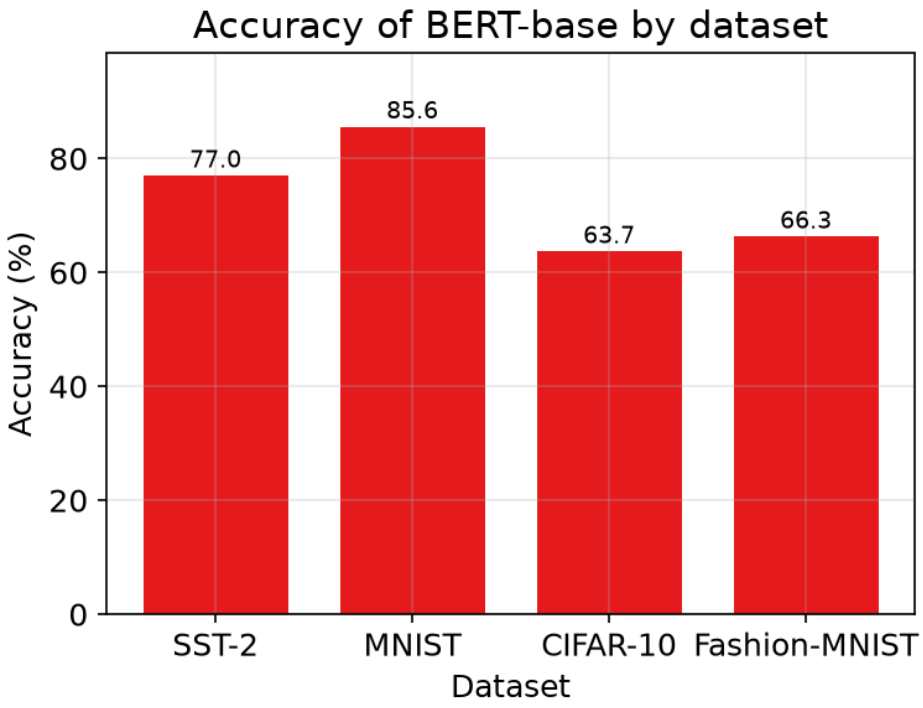


Figure 3: Accuracy of BERT-base by dataset for the models in this experiment.

BERT-base reaches 85.6% accuracy on SST-2. As shown in Figure 3, on SST-2, BERT-base obtains a accuracy of 85.6%. BERT-base scores 28.3 points higher on MNIST than on SST-2.

3 Discussion

On AG News, Naive Bayes obtains an accuracy of 89.0%. The differences between the models are discussed in the next section.